VCE CHEMISTRY · PRACTICE EXAM

Chemistry · Harder paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ **Date** _____

110 marks

reading 15 min · writing 150 min

State average on these questions

45 / 110 (41%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. v = sqrt(2gh) = 9.9 m/s.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
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5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the hardest questions. Questions on content that isn't in the current study design have been left out.

Sources

- 2022 Chemistry exam (PDF) · examiners' report *(previous study design)*
- 2023 Chemistry exam (PDF) · examiners' report *(previous study design)*
- 2024 Chemistry exam (PDF) · examiners' report
- 2025 Chemistry exam (PDF) · examiners' report

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–D. One mark each.

Question 1 · 1 mark

2022 VCAA exam · Section A · Question 12 · previous study design · open in the official exam PDF (p. 6) ↗

Why acidic conditions stop enzymes working

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 2 · 1 mark

2022 VCAA exam · Section A · Question 13 · previous study design · open in the official exam PDF (p. 6) ↗

Which electrolyte gives a gas at the cathode with inert electrodes

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 3 · 1 mark

2022 VCAA exam · Section A · Question 18 · previous study design · open in the official exam PDF (p. 7) ↗

Choosing electrodes and solutions for a working Sn^{4+}/Sn^{2+} and Ag^{+}/Ag cell

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 4 · 1 mark

2022 VCAA exam · Section A · Question 20 · previous study design · open in the official exam PDF (p. 8) ↗

Zinc placed in a mixed Co^{2+}/Mn^{2+} solution (no external circuit)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 5 · 1 mark

2022 VCAA exam · Section A · Question 28 · previous study design · open in the official exam PDF (p. 11) ↗

Matching a ^{13}C spectrum (five signals, one near 140 ppm) to an alcohol

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 6 · 1 mark

2022 VCAA exam · Section A · Question 30 · previous study design · open in the official exam PDF (p. 13) ↗

Ranking unfamiliar half-equations from reaction data and choosing the highest-EMF cell

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 7 · 1 mark

2023 VCAA exam · Section A · Question 2 · previous study design · open in the official exam PDF (p. 2) ↗

Which statement about fuel cells and galvanic cells is true

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 8 · 1 mark

2023 VCAA exam · Section A · Question 6 · previous study design · open in the official exam PDF (p. 3) ↗

Which statements describe what HPLC can do

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 9 · 1 mark

2023 VCAA exam · Section A · Question 10 · previous study design · open in the official exam PDF (p. 4) ↗

What always increases the equilibrium yield at constant temperature

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 10 · 1 mark

2023 VCAA exam · Section A · Question 14 · previous study design · open in the official exam PDF (p. 6) ↗

Effect of a temperature rise on the forward and reverse rates of an endothermic equilibrium

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 11 · 1 mark

2023 VCAA exam · Section A · Question 15 · previous study design · open in the official exam PDF (p. 6) ↗

Comparing petrol-powered and hydrogen-fuel-cell drones

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 12 · 1 mark

2023 VCAA exam · Section A · Question 16 · previous study design · open in the official exam PDF (p. 6) ↗

Number of ^{13}C NMR signals for a branched ester

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 13 · 1 mark

2023 VCAA exam · Section A · Question 18 · previous study design · open in the official exam PDF (p. 7) ↗

Reading an HPLC calibration curve and correcting for dilution

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 14 · 1 mark

2023 VCAA exam · Section A · Question 22 · previous study design · open in the official exam PDF (p. 8) ↗

Greenhouse gases and electrons per mole for methane vs methanol fuel cells

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 15 · 1 mark

2023 VCAA exam · Section A · Question 25 · previous study design · open in the official exam PDF (p. 9) ↗

Which species react at each electrode of a lead–acid cell during discharge and recharge

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 16 · 1 mark

2024 VCAA exam · Section A · Question 2 · open in the official exam PDF (p. 2) ↗

Which statement about glucose oxidation and combustion is correct

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 17 · 1 mark

2024 VCAA exam · Section A · Question 6 · open in the official exam PDF (p. 4) ↗

Galvanic cell with an iodate/iodine half-cell and an alkaline oxygen half-cell

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 18 · 1 mark

2024 VCAA exam · Section A · Question 9 · open in the official exam PDF (p. 5) ↗

Moles of OH^- consumed per mole of ethanol at the anode of an alkaline fuel cell

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 19 · 1 mark

2024 VCAA exam · Section A · Question 12 · open in the official exam PDF (p. 5) ↗

Why combustion of butane is treated as irreversible

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 20 · 1 mark

2024 VCAA exam · Section A · Question 17 · open in the official exam PDF (p. 7) ↗

Electrolytic refining of blister copper

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 21 · 1 mark

2024 VCAA exam · Section A · Question 19 · open in the official exam PDF (p. 8) ↗

Comparing cyclohexane and benzene

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 22 · 1 mark

2024 VCAA exam · Section A · Question 21 · open in the official exam PDF (p. 8) ↗

Skeletal structure of 3-chlorobutyl butanoate

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 23 · 1 mark

2024 VCAA exam · Section A · Question 24 · open in the official exam PDF (p. 9) ↗

Redox titration volume: permanganate with oxalic acid

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 24 · 1 mark

2025 VCAA exam · Section A · Question 2 · open in the official exam PDF (p. 2) ↗

Energy content per 100 g of a snack bar from its macronutrients

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 25 · 1 mark

2025 VCAA exam · Section A · Question 7 · open in the official exam PDF (p. 5) ↗

What migrates through the polymer/gel electrolyte of a flexible cell

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 26 · 1 mark

2025 VCAA exam · Section A · Question 8 · open in the official exam PDF (p. 5) ↗

Which metal–air cell (metal, electrolyte) gives the highest potential difference

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 27 · 1 mark

2025 VCAA exam · Section A · Question 18 · open in the official exam PDF (p. 9) ↗

What happens in artificial photosynthesis

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 28 · 1 mark

2025 VCAA exam · Section A · Question 19 · open in the official exam PDF (p. 10) ↗

Ordering lab glassware by resolution

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 29 · 1 mark

2025 VCAA exam · Section A · Question 27 · open in the official exam PDF (p. 13) ↗

A lab test that distinguishes an aldehyde from a tertiary alcohol

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Section B – Short- and extended-answer questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (10 marks)

2022 VCAA exam · Section B · Question 5 · previous study design · open in the official exam PDF (p. 22) ↗

Identifying a chlorinated ester from ^{13}C NMR, IR, mass spectrum (Cl isotopes) and ^{1}H NMR

a. A carbon environment that gives a peak at 168 ppm (1 mark)

b. Functional group at 168 ppm consistent with the IR, justified (2 marks)

c.i. Why there is a peak at m/z 110 (1 mark)

c.ii. How the mass-spectrum peaks relate to the molecular mass of 108.5 (2 marks)

d. What two singlets in the ^1H NMR tell you (2 marks)

e. Draw Molecule J (2 marks)

Question 2 (7 marks)

2023 VCAA exam · Section B · Question 6 · previous study design · open in the official exam PDF (p. 24) ↗

An experimental rechargeable flow cell: oxidising agent, recharge half-equation and electrode polarity

a. Energy transformation during discharge (1 mark)

b. The oxidising agent on discharge, using oxidation numbers (2 marks)

c.i. Half-equation in the organic half-cell during recharge (1 mark)

c.ii. Polarity of that electrode during recharge (1 mark)

c.iii. How electrode polarity is set up during recharge (2 marks)

Question 3 (8 marks)

2023 VCAA exam · Section B · Question 9 · previous study design · open in the official exam PDF (p. 34) ↗

Extended response: electrolytic refining of copper (Zn, Pb, Au impurities), then using Faraday's law

a. Discuss how electrolysis purifies copper, defining electrolysis and explaining the fate of Au, Zn and Pb (5 marks)

b. Explain how F and Faraday's first law give the amount of Br_2 from molten KBr (3 marks)

Question 4 (12 marks)

2024 VCAA exam · Section B · Question 8 · open in the official exam PDF (p. 30) ↗

Student investigation of fermentation rate: two gas-measuring methods, data resolution, hypothesis, limitations

a.i. Equation for producing bioethanol from glucose (1 mark)

a.ii. Why boiling water isn't used (1 mark)

b.i. Moles of gas from the mass loss (Method A) (1 mark)

b.ii. Moles of gas from the volume collected at SLC (Method B) (1 mark)

b.iii. An assumption behind the part b.ii calculation (1 mark)

b.iv. Compare the resolution of the data in the two methods (2 marks)

c.i. The independent variable (1 mark)

c.ii. Does the graph support the hypothesis? (2 marks)

d.i. A limitation of Method B (different from part b) (1 mark)

d.ii. Which poster section reports limitations (1 mark)

Question 5 (17 marks)

2025 VCAA exam · Section B · Question 4 · open in the official exam PDF (p. 23) ↗

Organic synthesis (haloalkane, amine), boiling points of acids vs amines, then identifying C_4H_9ClO from spectra

a.i. Equation for making C_3H_7Cl from propane, with conditions (2 marks)

a.ii. Name two reactants that form propan-1-amine in one substitution step (2 marks)

b.i. Why carboxylic acids boil higher than the matching amines (2 marks)

b.ii. Whether the IR spectrum is of an amine or a carboxylic acid (1 mark)

c.i. m/z of the base peak (1 mark)

c.ii. Why every isomer has a peak at m/z 108 (1 mark)

c.iii. Formula of the species at m/z 95 (1 mark)

d. Three deductions from the ^{1}H NMR data (3 marks)

e. What the ^{13}C NMR shows (2 marks)

f.i. Skeletal diagram of Compound X (1 mark)

f.ii. IUPAC name of Compound X (1 mark)

Question 6 (13 marks)

2025 VCAA exam · Section B · Question 5 · open in the official exam PDF (p. 28) ↗

Quinine in tonic water: HPLC (retention time, calibration), functional groups, pH, chirality, competitive inhibition

a.i. How retention time confirms quinine is present (2 marks)

a.ii. Why standards must be run under controlled conditions (2 marks)

a.iii. Why the curve can't be used for a 130 mg L^{-1} sample (1 mark)

b.i. Highest-priority functional group in quinine (IUPAC) (1 mark)

b.ii. Which structure occurs at low pH, justified (2 marks)

c.i. Why only one isomer of quinine is effective, using “chiral centre” (3 marks)

c.ii. Why quinine inhibits better at pH 4.7 than at higher pH (2 marks)

Question 7 (14 marks)

2025 VCAA exam · Section B · Question 6 · open in the official exam PDF (p. 32) ↗

Student investigation: redox titration of oxalate in spinach before and after boiling, evaluated against a hypothesis

a. Methodology used to develop the hypothesis (1 mark)

b. The end-point observation in a permanganate titration (1 mark)

c.i. Why 50 °C is preferred, using collision theory (1 mark)

c.ii. Why the experiment might not be valid at room temperature (1 mark)

d. Average titre (1 mark)

e. Mass of oxalate in the 15.15 g uncooked sample (4 marks)

f. Percentage change in oxalate after boiling (1 mark)

g. Why insufficient rinsing would affect the result (2 marks)

h. Evaluate the hypothesis against the result (2 marks)

Marking guide follows

Marking guide

Mark yourself only after you've finished. Tick the boxes you earned, enter your marks and compare with how Victoria went on each question.

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	**A**	Excess H^+ protonates basic R groups (e.g. $–NH_2$ becomes $–NH_3^+$). This changes the charges and the interactions that hold the tertiary structure and active site in shape.	42% correct A 42% · B 19% · C 12% · D 27%
☐	2	**A**	For KI(aq), water is a stronger oxidant than K^+, so H_2 forms at the cathode. The molten salts give metals, and $CuSO_4$(aq) gives copper.	29% correct A 29% · B 39% · C 10% · D 21%
☐	3	**C**	The Sn^{4+}/Sn^{2+} half-cell needs both ions and an inert (graphite) electrode, because a Sn electrode would bring in the Sn^{2+}/Sn couple. The Ag^+/Ag half-cell needs an Ag electrode.	44% correct A 22% · B 13% · C 44% · D 20%
☐	4	**B**	Zn reduces Co^{2+} directly in the beaker. There is no circuit, so no voltage is produced and the chemical energy is released as heat.	34% correct A 34% · B 34% · C 15% · D 17%
☐	5	**D**	Five signals means five carbon environments, and a signal near 110–140 ppm means C=C. Only $CH_2{=}CHCH_2CH_2CH_2OH$ has both. Some candidates have five environments but no C=C, or C=C but six environments.	42% correct A 14% · B 32% · C 12% · D 42%
☐	6	**A**	Use the reaction information to build your own series: HBrO > BrO_3^- > IO_3^- > $HCrO_4^-$. The largest gap, and so the highest EMF, is HBrO/Br_2 paired with $HCrO_4^-/Cr^{3+}$.	34% correct A 34% · B 24% · C 23% · D 18%
☐	7	**A**	Both release some energy as heat. Galvanic cells can have inert electrodes too, porous electrodes are not what separates them, and electrons never flow through the electrolyte.	40% correct A 40% · B 5% · C 42% · D 13%
☐	8	**C**	HPLC is qualitative (retention times compared with standards) and quantitative (peak areas against a calibration curve). "Triplets" is NMR splitting, which has nothing to do with HPLC.	43% correct A 16% · B 30% · C 43% · D 10%
☐	9	**A**	Only a larger equilibrium-expression value means proportionally more product. Adding reactant raises the amount of product but not the percentage yield. A catalyst changes only the rate, and pressure depends on the moles of gas.	26% correct A 26% · B 47% · C 18% · D 8%
☐	10	**C**	Raising T speeds up both reactions immediately, because more collisions exceed Ea. The endothermic forward reaction is favoured, so it speeds up more. Picking "forward up, reverse down" confuses rate with equilibrium position.	44% correct A 43% · B 1% · C 44% · D 12%
☐	11	**A**	Both release H_2O(g), which is a greenhouse gas, and petrol also releases CO_2. The energy transformations differ (combustion vs electrochemical), and both produce heat.	41% correct A 41% · B 12% · C 35% · D 12%
☐	12	**C**	Draw it out as \(\text{CH}_3\text{CH}_2\text{CH(CH}_3)\text{COOCH(CH}_3)_2\). The two methyls on the isopropyl group are equivalent, which gives 7 carbon environments.	43% correct A 21% · B 23% · C 43% · D 12%
☐	13	**C**	The graph gives 18 ng mL^{-1} in the diluted sample. The sample was diluted 1000× (1 mL to 1 L), so the original is 18 µg mL^{-1}. Forgetting the dilution was the popular wrong answer.	37% correct A 8% · B 45% · C 37% · D 9%
☐	14	**B**	Both give 1 CO_2 + 2 H_2O per mole of fuel. The anode half-equations release 8 e^- (CH_4) vs 6 e^- (CH_3OH), so the electron counts differ.	41% correct A 5% · B 41% · C 17% · D 37%
☐	15	**C**	Write both half-equations in each direction. On discharge, H^+ is consumed at the cathode (PbO_2 reduction). HSO_4^- is consumed at both electrodes on discharge and produced at both on recharge.	44% correct A 28% · B 16% · C 44% · D 11%

☐	16	**B**	From the data book, glucose ≈ 16 kJ g^{-1} and H_2 ≈ 143 kJ g^{-1}, so glucose gives less energy per gram. The equation offered is photosynthesis, and oxygen does not “combust”.	40% correct A 20% · B 40% · C 15% · D 25%
☐	17	**C**	IO_3^- (+1.20 V) is the stronger oxidant, so the O_2/OH^- half-cell is the anode (negative). There \(4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4e^-\), so $[OH^-]$ falls at the negative electrode.	43% correct A 11% · B 29% · C 43% · D 16%
☐	18	**D**	Balance in acid first (\(\text{C}_2\text{H}_5\text{OH} + 3\text{H}_2\text{O} \rightarrow 2\text{CO}_2 + 12\text{H}^+ + 12e^-\)), then add 12 OH^- to both sides. That gives 12 mol OH^-.	29% correct A 20% · B 27% · C 23% · D 29%
☐	19	**C**	The reverse reaction is so slow (a vanishingly small chance of a successful collision) that it can be ignored. Being exothermic does not make a reaction irreversible.	32% correct A 16% · B 20% · C 32% · D 32%
☐	20	**D**	More reactive impurities (Fe, Ni) are oxidised at the anode in place of some Cu, but only Cu^{2+} is reduced at the cathode. So $[Cu^{2+}]$ in the electrolyte falls during electrolysis. The impure copper is the anode.	38% correct A 20% · B 27% · C 15% · D 38%
☐	21	**A**	Both formulas have non-cyclic structural isomers (hexenes for C_6H_{12}, hexadiynes for C_6H_6). Their C–C bond strengths differ, they belong to different homologous series, and every carbon has 4 valence electrons.	34% correct A 34% · B 14% · C 22% · D 30%
☐	22	**D**	The “3-chlorobutyl” part is the alcohol side ($O–CH_2CH_2CHClCH_3$, numbered from the O). The “butanoate” part is the 4-carbon acid side including C=O. Most wrong answers put the Cl on the acid chain.	42% correct A 33% · B 5% · C 20% · D 42%
☐	23	**D**	\(\text{MnO}_4^-\) takes 5 e^- and oxalic acid gives 2 e^-, so the ratio is 2:5. $n(MnO_4^-)$ = 0.00100 mol gives n(oxalic) = 0.00250 mol, which is 25.0 mL. Assuming 1:1 gives 10 mL (the popular trap).	26% correct A 14% · B 35% · C 24% · D 26%
☐	24	**D**	\((16 \times 45) + (17 \times 10) + (37 \times 10) = 1260\ \text{kJ}\) per 65 g, which is \(1938\ \text{kJ} = 1.9 \times 10^6\ \text{J}\) per 100 g. The common error was a kJ/J mix-up.	34% correct A 18% · B 35% · C 13% · D 34%
☐	25	**B**	Electrons go through the external wire and never through the electrolyte. Through the electrolyte, metal ions made at the anode migrate towards the cathode. 60% chose the electron option.	24% correct A 60% · B 24% · C 9% · D 6%
☐	26	**A**	The oxygen half-cell is +1.23 V in acid but only +0.40 V in alkali. Mg–air in acid: 1.23 − (−2.37) = 3.60 V, which beats Na–air in alkali (3.1 V), the popular choice.	29% correct A 29% · B 7% · C 53% · D 11%
☐	27	**A**	Sunlight drives the splitting of water: water is oxidised (to O_2) and H_2 is produced. Reduction happens at the cathode, not the anode, and no external power supply is used.	41% correct A 41% · B 21% · C 15% · D 23%
☐	28	**C**	Higher resolution means finer graduations. A beaker is the coarsest, then the 50 mL cylinder, then the 10 mL cylinder, and the 3 mL graduated pipette is the finest.	34% correct A 19% · B 21% · C 34% · D 25%
☐	29	**A**	Geranial (an aldehyde) is oxidised by acidified dichromate, turning it orange → green. Linalool is a tertiary alcohol and does not react. Both have C=C, so the bromine test cannot tell them apart.	41% correct A 41% · B 21% · C 13% · D 24%

Multiple-choice score: ☐ **/ 29** **(state average 10.7)**

Section B – Short- and extended-answer questions

Question 1 (10 marks) · state average 3.9 / 10 (39%)

2022 VCAA exam · Section B · Question 5

a. A carbon environment that gives a peak at 168 ppm

- [ ] Carboxylic acid or ester carbonyl carbon

Answer / notes: An ester (or acid) C=O carbon.

State average 0.9 / 1 (90%); 89% scored full marks. Mark distribution: 0: 11% · 1: 89%

Your mark: ____ **/ 1**

b. Functional group at 168 ppm consistent with the IR, justified

- [ ] Ester
- [ ] No broad O–H (acid) absorption at 2500–3500 cm^{-1}, so it is not a carboxylic acid

Answer / notes: Ester. The IR has a C=O but no broad acid O–H band.

State average 0.8 / 2 (40%); 20% scored full marks. Mark distribution: 0: 39% · 1: 41% · 2: 20%

Your mark: ____ **/ 2**

c.i. Why there is a peak at m/z 110

- [ ] Molecules containing the ^{37}Cl isotope

Answer / notes: The molecular ion containing ^{37}Cl. The ^{35}Cl version is at 108.

State average 0.1 / 1 (10%); 15% scored full marks. Mark distribution: 0: 85% · 1: 15%

Your mark: ____ **/ 1**

c.ii. How the mass-spectrum peaks relate to the molecular mass of 108.5

- [ ] The peaks at 108 and 110 are both molecular (parent) ions
- [ ] The molar mass 108.5 is the weighted average of these by relative abundance (≈ 3:1)

Answer / notes: Both 108 and 110 are parent ions, and 108.5 is their abundance-weighted average. 97% of the state scored zero.

State average 0.0 / 2 (2%); 1% scored full marks. Mark distribution: 0: 97% · 1: 2% · 2: 1%

Your mark: ____ **/ 2**

d. What two singlets in the ^{1}H NMR tell you

- [ ] Two hydrogen environments
- [ ] No H on adjacent carbons (singlets), or an H ratio of 2:3

Answer / notes: Two H environments (a 2:3 ratio), neither with neighbouring H atoms.

State average 1.3 / 2 (65%); 45% scored full marks. Mark distribution: 0: 17% · 1: 38% · 2: 45%

Your mark: ____ **/ 2**

e. Draw Molecule J

- [ ] Any valid structure of $C_3H_5O_2Cl$
- [ ] Methyl chloroethanoate ($ClCH_2COOCH_3$) or chloromethyl ethanoate (CH_3COOCH_2Cl)

Answer / notes: $ClCH_2COOCH_3$: an ester with isolated CH_2 and CH_3 groups.

State average 0.8 / 2 (40%); 35% scored full marks. Mark distribution: 0: 55% · 1: 10% · 2: 35%

Your mark: ____ **/ 2**

Question 2 (7 marks) · state average 3.1 / 7 (44%)

2023 VCAA exam · Section B · Question 6

a. Energy transformation during discharge

- [ ] Chemical to electrical energy

Answer / notes: Chemical to electrical.

State average 0.8 / 1 (80%); 80% scored full marks. Mark distribution: 0: 20% · 1: 80%

Your mark: ____ **/ 1**

b. The oxidising agent on discharge, using oxidation numbers

- [ ] $Fe_2(SO_4)_3$ (Fe^{3+})
- [ ] Fe goes from +3 to +2 (reduced), so it is the oxidant

Answer / notes: Fe^{3+}, since its oxidation number falls from +3 to +2.

State average 0.9 / 2 (45%); 30% scored full marks. Mark distribution: 0: 40% · 1: 30% · 2: 30%

Your mark: ☐ **/ 2**

c.i. Half-equation in the organic half-cell during recharge

- [ ] \(\text{C}_{14}\text{H}_8\text{O}_8\text{S}_2 + 2\text{H}^+ + 2e^- \rightarrow \text{C}_{14}\text{H}_{10}\text{O}_8\text{S}_2\)

Answer / notes: \(\text{C}_{14}\text{H}_8\text{O}_8\text{S}_2 + 2\text{H}^+ + 2e^- \rightarrow \text{C}_{14}\text{H}_{10}\text{O}_8\text{S}_2\)

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: ☐ **/ 1**

c.ii. Polarity of that electrode during recharge

- [ ] Negative

Answer / notes: Negative.

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: ☐ **/ 1**

c.iii. How electrode polarity is set up during recharge

- [ ] Polarity stays the same as on discharge (set by the relative strengths of the oxidant and reductant)
- [ ] The power supply's negative terminal is connected to the negative electrode, forcing electrons in so that the reactions reverse (negative becomes the cathode)

Answer / notes: The polarity doesn't flip. You connect negative to negative, and the supply pushes electrons the opposite way, so the anode and cathode swap. 70% scored zero.

State average 0.4 / 2 (20%); 10% scored full marks. Mark distribution: 0: 70% · 1: 20% · 2: 10%

Your mark: ☐ **/ 2**

Question 3 (8 marks) · state average 2.4 / 8 (30%)

2023 VCAA exam · Section B · Question 9

a. Discuss how electrolysis purifies copper, defining electrolysis and explaining the fate of Au, Zn and Pb

- [ ] Definition: an electric current drives a non-spontaneous redox reaction
- [ ] At the impure anode, Zn and Pb (stronger reductants than Cu) are oxidised to Zn^{2+} and Pb^{2+}
- [ ] Au is a weaker reductant than Cu, so it is not oxidised and falls to the bottom as anode sludge
- [ ] Cu is oxidised at the anode, and Cu^{2+} (the strongest oxidant present) is reduced to pure Cu at the cathode
- [ ] Zn^{2+} is too weak an oxidant to be reduced, so it stays in solution (Pb^{2+} forms insoluble $PbSO_4$ with the sulfate)

Answer / notes: Go through every species using the electrochemical series. The state average was 1.2/5 because most answers missed the fate of each impurity.

State average 1.2 / 5 (24%); 1% scored full marks. Mark distribution: 0: 30% · 1: 40% · 2: 20% · 3: 10% · 4: 4% · 5: 1%

Your mark: ☐ **/ 5**

b. Explain how F and Faraday's first law give the amount of Br_2 from molten KBr

- [ ] Q = It
- [ ] $n(e^-) = Q/F$
- [ ] $n(Br_2) = ½\ n(e^-)$, from \(2\text{Br}^- \rightarrow \text{Br}_2 + 2e^-\)

Answer / notes: Charge from Q = It, moles of electrons from Q/F, then use the 2 e^- per Br_2 ratio.

State average 1.2 / 3 (40%); 20% scored full marks. Mark distribution: 0: 40% · 1: 20% · 2: 20% · 3: 20%

Your mark: ☐ **/ 3**

Question 4 (12 marks) · state average 5.2 / 12 (43%)

2024 VCAA exam · Section B · Question 8

a.i. Equation for producing bioethanol from glucose

- [ ] \(\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2\)

Answer / notes: \(\text{C}_6\text{H}_{12}\text{O}_6(aq) \rightarrow 2\text{C}_2\text{H}_5\text{OH}(aq) + 2\text{CO}_2(g)\)

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: ☐ **/ 1**

a.ii. Why boiling water isn't used

- [ ] It would denature the enzymes (in the yeast)

Answer / notes: High temperature denatures the enzymes. "Kills the yeast" didn't score.

State average 0.4 / 1 (40%); 37% scored full marks. Mark distribution: 0: 63% · 1: 37%

Your mark: ☐ **/ 1**

b.i. Moles of gas from the mass loss (Method A)

- [ ] 0.1/44.0 = 0.0023 mol

Answer / notes: 0.1 g ÷ 44.0 = 2.3×10^{-3} mol.

State average 0.6 / 1 (60%); 61% scored full marks. Mark distribution: 0: 39% · 1: 61%

Your mark: ☐ **/ 1**

b.ii. Moles of gas from the volume collected at SLC (Method B)

- [ ] (32.5 − 0.5) mL = 0.0320 L, ÷ 24.8 = 1.29×10^{-3} mol

Answer / notes: Subtract the initial reading and convert to litres: 1.29×10^{-3} mol.

State average 0.5 / 1 (50%); 43% scored full marks. Mark distribution: 0: 57% · 1: 43%

Your mark: ☐ **/ 1**

b.iii. An assumption behind the part b.ii calculation

- [ ] A valid assumption (the gas is ideal, CO_2 doesn't dissolve in the water, the gas is at 25 °C and 100 kPa, or it is only CO_2)

Answer / notes: For example, no CO_2 dissolves in the water. "Done at SLC" wasn't enough because the flask was at 35 °C.

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: ☐ **/ 1**

b.iv. Compare the resolution of the data in the two methods

- [ ] Balance (Method A): 0.1 g
- [ ] Burette (Method B): 1 mL (as given in the report)

Answer / notes: Give each instrument's smallest reading step with units. Only 1% got both marks.

State average 0.2 / 2 (10%); 1% scored full marks. Mark distribution: 0: 86% · 1: 13% · 2: 1%

Your mark: ☐ **/ 2**

c.i. The independent variable

- [ ] Concentration of the glucose solution

Answer / notes: Glucose concentration.

State average 0.7 / 1 (70%); 69% scored full marks. Mark distribution: 0: 31% · 1: 69%

Your mark: ☐ **/ 1**

c.ii. Does the graph support the hypothesis?

- [ ] No, not supported
- [ ] The graph shows the same gradient / volume for both concentrations over the first ~15 minutes

Answer / notes: No. Both lines have the same slope for the first 15 minutes, so the rate didn't increase. Refer to the graph explicitly.

State average 0.7 / 2 (35%); 17% scored full marks. Mark distribution: 0: 52% · 1: 30% · 2: 17%

Your mark: ____ **/ 2**

d.i. A limitation of Method B (different from part b)

- [ ] A valid limitation (gas dissolves in water, gas escapes before the stopper is on, the burette can't hold 500 mL, the gas isn't at 25 °C)

Answer / notes: For example, the 50 mL burette can't collect the ~500 mL shown on the graph.

State average 0.4 / 1 (40%); 33% scored full marks. Mark distribution: 0: 67% · 1: 33%

Your mark: ____ **/ 1**

d.ii. Which poster section reports limitations

- [ ] Discussion

Answer / notes: The discussion.

State average 0.9 / 1 (90%); 87% scored full marks. Mark distribution: 0: 13% · 1: 87%

Your mark: ____ **/ 1**

Question 5 (17 marks) · state average 7.4 / 17 (44%)

2025 VCAA exam · Section B · Question 4

a.i. Equation for making C_3H_7Cl from propane, with conditions

- [ ] $C_3H_8 + Cl_2 \rightarrow C_3H_7Cl + HCl$
- [ ] UV light

Answer / notes: \(\text{C}_3\text{H}_8 + \text{Cl}_2 \xrightarrow{\text{UV}} \text{C}_3\text{H}_7\text{Cl} + \text{HCl}\)

State average 1.3 / 2 (65%); 48% scored full marks. Mark distribution: 0: 25% · 1: 27% · 2: 48%

Your mark: ____ **/ 2**

a.ii. Name two reactants that form propan-1-amine in one substitution step

- [ ] 1-chloropropane (a primary 3-carbon haloalkane)
- [ ] Ammonia

Answer / notes: 1-chloropropane and ammonia. Give names, not formulas, when the question asks for names.

State average 0.6 / 2 (30%); 17% scored full marks. Mark distribution: 0: 56% · 1: 27% · 2: 17%

Your mark: ____ **/ 2**

b.i. Why carboxylic acids boil higher than the matching amines

- [ ] Acids form more or stronger hydrogen bonds (O–H, C=O, dimers) than amines
- [ ] So more energy is needed to separate the molecules

Answer / notes: COOH groups hydrogen-bond more strongly (they even form dimers) than N–H does, so more energy is needed to boil them. Talking about breaking covalent bonds scored zero.

State average 1.0 / 2 (50%); 29% scored full marks. Mark distribution: 0: 35% · 1: 37% · 2: 29%

Your mark: ____ **/ 2**

b.ii. Whether the IR spectrum is of an amine or a carboxylic acid

- [ ] Amine

Answer / notes: Amine: there is no C=O and no broad acid O–H.

State average 0.8 / 1 (80%); 77% scored full marks. Mark distribution: 0: 23% · 1: 77%

Your mark: ____ **/ 1**

c.i. m/z of the base peak

- [ ] 59

Answer / notes: 59.

State average 0.9 / 1 (90%); 86% scored full marks. Mark distribution: 0: 14% · 1: 86%

Your mark: ____ **/ 1**

c.ii. Why every isomer has a peak at m/z 108

- [ ] They all have the same molecular formula, so the molecular ion containing ^{35}Cl is at 108

Answer / notes: Same formula, so the same ^{35}Cl molecular ion. The molar mass is 108.5, not 108.

State average 0.4 / 1 (40%); 41% scored full marks. Mark distribution: 0: 59% · 1: 41%

Your mark: ____ **/ 1**

c.iii. Formula of the species at m/z 95

- [ ] \([\text{C}_3\text{H}_6{}^{37}\text{ClO}]^+\)

Answer / notes: \([\text{C}_3\text{H}_6{}^{37}\text{ClO}]^+\): the molecular ion with ^{37}Cl minus CH_3. Label the isotope and the + charge (only 10% did).

State average 0.1 / 1 (10%); 10% scored full marks. Mark distribution: 0: 90% · 1: 10%

Your mark: ____ **/ 1**

d. Three deductions from the ^{1}H NMR data

- [ ] Three hydrogen environments
- [ ] All singlets, so no H on adjacent carbons
- [ ] Area ratio 1:2:6 (e.g. two equivalent CH_3 groups)

Answer / notes: Three environments, all isolated (singlets), in a 1:2:6 ratio. Use the data, not guesses from chemical shifts.

State average 0.8 / 3 (27%); 5% scored full marks. Mark distribution: 0: 46% · 1: 32% · 2: 17% · 3: 5%

Your mark: ____ **/ 3**

e. What the ^{13}C NMR shows

- [ ] Three carbon environments
- [ ] So there is symmetry (two equivalent carbons among the four)

Answer / notes: Three environments for four carbons, so two are equivalent.

State average 1.0 / 2 (50%); 19% scored full marks. Mark distribution: 0: 17% · 1: 64% · 2: 19%

Your mark: ____ **/ 2**

f.i. Skeletal diagram of Compound X

- [ ] A correct skeletal structure (1-chloro-2-methylpropan-2-ol or 2-chloro-2-methylpropan-1-ol)

Answer / notes: A central carbon bearing two CH_3 groups, an OH and a CH_2Cl, drawn as a skeletal structure (only 15% did).

State average 0.2 / 1 (20%); 15% scored full marks. Mark distribution: 0: 85% · 1: 15%

Your mark: ____ **/ 1**

f.ii. IUPAC name of Compound X

- [ ] 1-chloro-2-methylpropan-2-ol (or the matching isomer name)

Answer / notes: 1-chloro-2-methylpropan-2-ol.

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: ____ **/ 1**

Question 6 (13 marks) · state average 5.6 / 13 (43%)

2025 VCAA exam · Section B · Question 5

a.i. How retention time confirms quinine is present

- [ ] Run a pure quinine standard under identical conditions to get its retention time
- [ ] A sample peak at the same retention time indicates quinine (or spiking increases that peak)

Answer / notes: Run a quinine standard under the same conditions. A matching retention time (1.415 min) supports its presence.

State average 0.7 / 2 (35%); 20% scored full marks. Mark distribution: 0: 52% · 1: 27% · 2: 20%

Your mark: ☐ **/ 2**

a.ii. Why standards must be run under controlled conditions

- [ ] Changing conditions change retention times and peak areas
- [ ] So the calibration curve (peak area vs concentration) would be invalid or inaccurate

Answer / notes: Peak area only maps to concentration if the flow rate, temperature, column and solvent are constant.

State average 0.4 / 2 (20%); 7% scored full marks. Mark distribution: 0: 68% · 1: 25% · 2: 7%

Your mark: ☐ **/ 2**

a.iii. Why the curve can't be used for a 130 mg L^{-1} sample

- [ ] It is outside the range of the standards (it would be extrapolation)

Answer / notes: It is outside the calibrated range. Dilute the sample first.

State average 0.8 / 1 (80%); 75% scored full marks. Mark distribution: 0: 25% · 1: 75%

Your mark: ☐ **/ 1**

b.i. Highest-priority functional group in quinine (IUPAC)

- [ ] Hydroxy (hydroxyl)

Answer / notes: Hydroxy (–OH).

State average 0.6 / 1 (60%); 57% scored full marks. Mark distribution: 0: 43% · 1: 57%

Your mark: ☐ **/ 1**

b.ii. Which structure occurs at low pH, justified

- [ ] Structure Q (the protonated one)
- [ ] At low pH the amine N atoms (bases) accept H^+

Answer / notes: Q. The amine nitrogens are protonated in acid.

State average 1.3 / 2 (65%); 46% scored full marks. Mark distribution: 0: 21% · 1: 33% · 2: 46%

Your mark: ☐ **/ 2**

c.i. Why only one isomer of quinine is effective, using "chiral centre"

- [ ] A chiral centre is a carbon with four different groups attached
- [ ] Four chiral centres give several isomers with different 3D shapes
- [ ] Only one has the complementary shape to bind the target enzyme's active site

Answer / notes: Different arrangements at the chiral centres change the 3D shape, and only one fits the active site.

State average 1.3 / 3 (43%); 14% scored full marks. Mark distribution: 0: 31% · 1: 28% · 2: 27% · 3: 14%

Your mark: ☐ **/ 3**

c.ii. Why quinine inhibits better at pH 4.7 than at higher pH

- [ ] At pH 4.7 the protonated form binds more strongly to the active site
- [ ] Because the charged ($–NH^+$) groups form ionic interactions with negative groups in the active site (or the shape fits better)

Answer / notes: Protonated quinine can form ionic bonds in the active site, so it competes more effectively with the substrate.

State average 0.5 / 2 (25%); 8% scored full marks. Mark distribution: 0: 60% · 1: 32% · 2: 8%

Your mark: ☐ **/ 2**

Question 7 (14 marks) · state average 6.3 / 14 (45%)

2025 VCAA exam · Section B · Question 6

a. Methodology used to develop the hypothesis

- [ ] Literature review

Answer / notes: Literature review (she researched it).

State average 0.5 / 1 (50%); 43% scored full marks. Mark distribution: 0: 57% · 1: 43%

Your mark: ____ **/ 1**

b. The end-point observation in a permanganate titration

- [ ] The first permanent purple colour (excess MnO_4^-)

Answer / notes: The first permanent purple tinge. “Pink” was rejected because Mn^{2+} is pale pink.

State average 0.2 / 1 (20%); 22% scored full marks. Mark distribution: 0: 78% · 1: 22%

Your mark: ____ **/ 1**

c.i. Why 50 °C is preferred, using collision theory

- [ ] A faster rate: more frequent collisions with energy ≥ Ea

Answer / notes: More particles have energy ≥ Ea, so the reaction is fast enough to titrate.

State average 0.9 / 1 (90%); 90% scored full marks. Mark distribution: 0: 10% · 1: 90%

Your mark: ____ **/ 1**

c.ii. Why the experiment might not be valid at room temperature

- [ ] The reaction would be too slow for a sharp end point (or it departs from the stated method)

Answer / notes: Too slow at room temperature, so the end point would be unreliable.

State average 0.7 / 1 (70%); 66% scored full marks. Mark distribution: 0: 34% · 1: 66%

Your mark: ____ **/ 1**

d. Average titre

- [ ] 15.42 mL (concordant titres only, excluding trial 1)

Answer / notes: (15.45 + 15.40 + 15.40)/3 = 15.42 mL. Leave out the rough first trial.

State average 0.7 / 1 (70%); 65% scored full marks. Mark distribution: 0: 35% · 1: 65%

Your mark: ____ **/ 1**

e. Mass of oxalate in the 15.15 g uncooked sample

- [ ] $n(MnO_4^-) = 3.08 \times 10^{-5}$ mol
- [ ] $n(C_2O_4^{2-})$ in the aliquot = 5/2 × that = 7.71×10^{-5} mol
- [ ] In 500 mL: × 20 = 1.54×10^{-3} mol
- [ ] Mass = 0.136 g (0.14 g)

Answer / notes: Mole ratio 5:2, then scale the 25.0 mL aliquot up to 500 mL: 0.14 g.

State average 1.7 / 4 (43%); 13% scored full marks. Mark distribution: 0: 36% · 1: 9% · 2: 18% · 3: 24% · 4: 13%

Your mark: ____ **/ 4**

f. Percentage change in oxalate after boiling

- [ ] ≈ 60% decrease

Answer / notes: (0.136 − 0.0545)/0.136 ≈ 60% decrease. 83% scored zero.

State average 0.2 / 1 (20%); 17% scored full marks. Mark distribution: 0: 83% · 1: 17%

Your mark: ____ **/ 1**

g. Why insufficient rinsing would affect the result

- [ ] Leftover cooking water on the leaves adds dissolved (soluble) oxalate to the analysis
- [ ] So a larger titre is needed, and the oxalate remaining is overestimated (the % removed is underestimated)

Answer / notes: Oxalate dissolved in the leftover water gets counted as if it were still in the leaves, so the titre is too big.

State average 0.5 / 2 (25%); 5% scored full marks. Mark distribution: 0: 61% · 1: 34% · 2: 5%

Your mark: ☐ **/ 2**

h. Evaluate the hypothesis against the result

- [ ] Refers to the 75% prediction in the hypothesis
- [ ] States that the 60% result is lower than 75%, so the hypothesis is not supported

Answer / notes: 60% < 75%, so the hypothesis (75% removal) is not supported.

State average 0.9 / 2 (45%); 36% scored full marks. Mark distribution: 0: 44% · 1: 20% · 2: 36%

Your mark: ☐ **/ 2**

Your result

Total: _____ **/ 110** **State average on these questions: 45 / 110 (41%)**

Topics to revise:

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